

$$\begin{array}{ccc}
 V & \xrightarrow{T} & W \\
 \varphi_{B_V} \downarrow & & \downarrow \varphi_{B_W} \\
 F^n & \xrightarrow{[T]_{\mathcal{B}_V, \mathcal{B}_W}} & F^m
 \end{array}$$